





WELCOME

Into the Box 2025: **The Future is Dynamic!**







Manager - Severed floor

Curt G

 @gratzc

 gratzc@compknowhow.com

 @gratzc

 @curtgratz

•My Outie

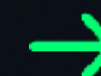
- Husband
- Dad
- Coach (CC and Track)
- Hike/Nature
- Duck Wrangler

•My Innie

- Co-Owner of CKH Consulting
- Less of a Werido then Luis
- BoxLang developer
- Polyglot
- Data Governance Nerd
- Member of the C&M Department

 Consulting







Vector Databases and the BoxLang/ CFML Developer

WTH is a Vector

Why would I want to use a Vector DB

AI and Vectors

DEMO BABY

GET STARTED



www.intothebox.org

Moonbeam

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7	2	7	7	4	1	7	1	4	0	2	7	9	6	3	2	0	2	9	3	3	8	8	5
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9	3	8	5	8	6	6	2	6	8	9	9	3	0	5	4	3	7	8	3	3	5	1	4
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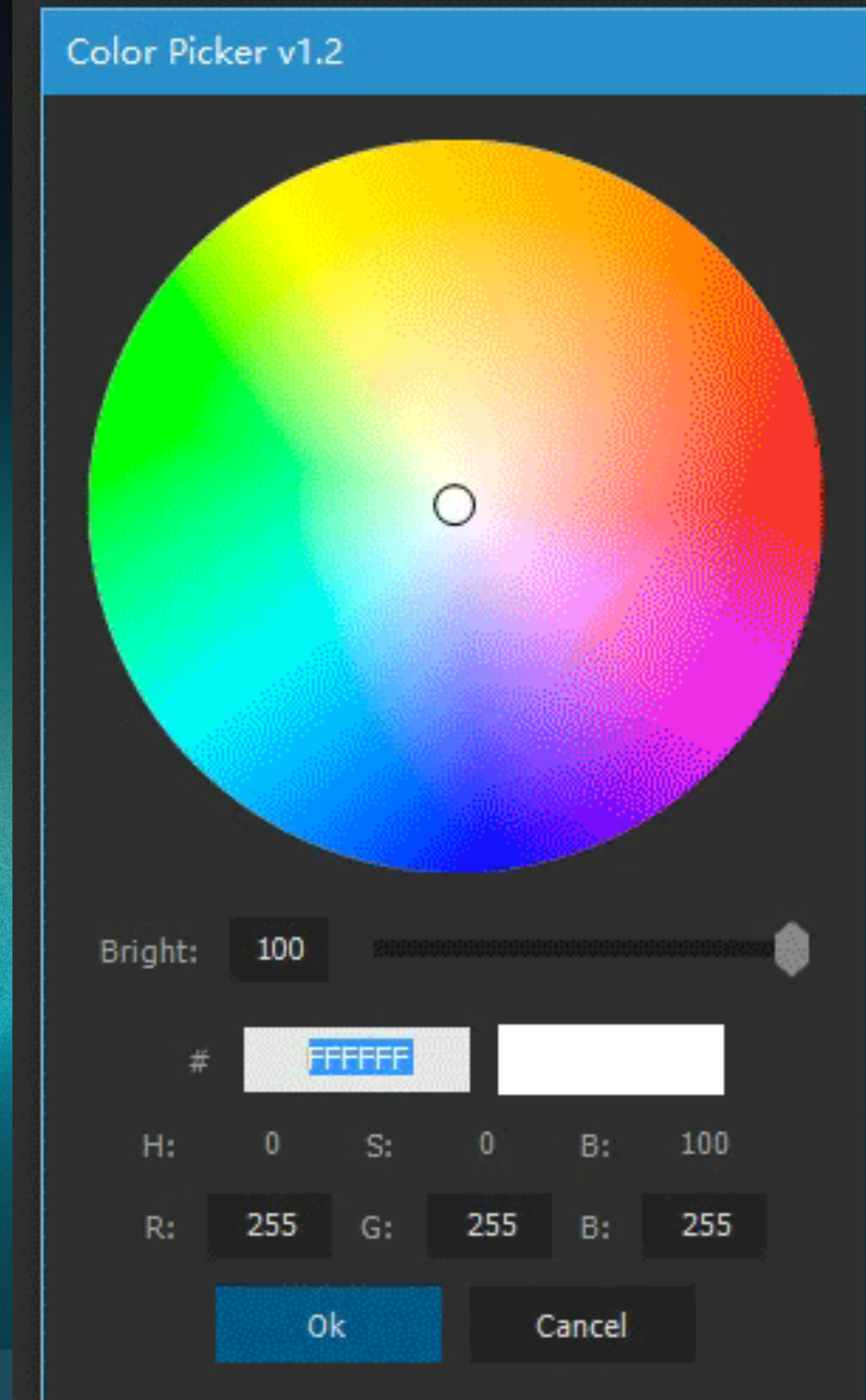
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Colors as Vectors

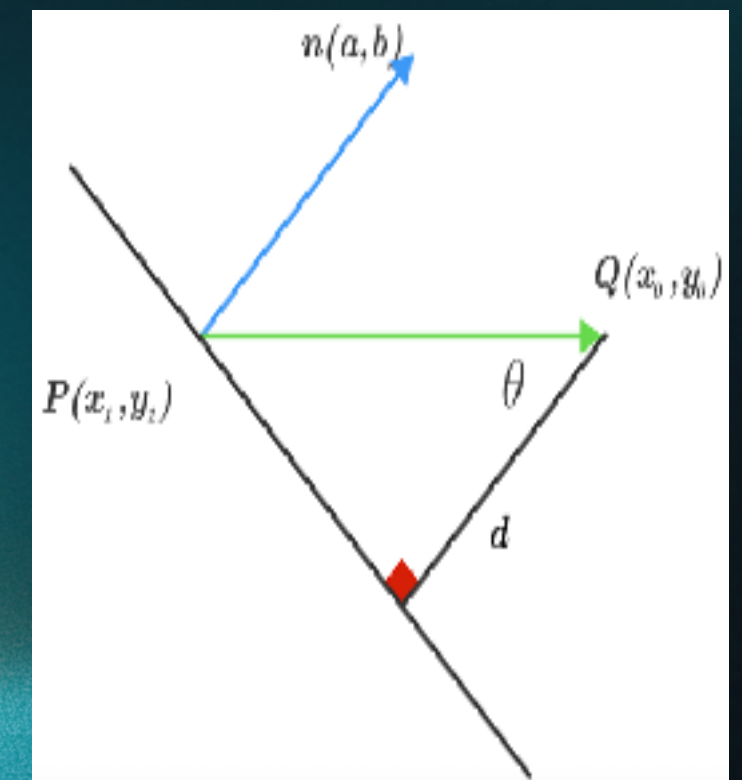
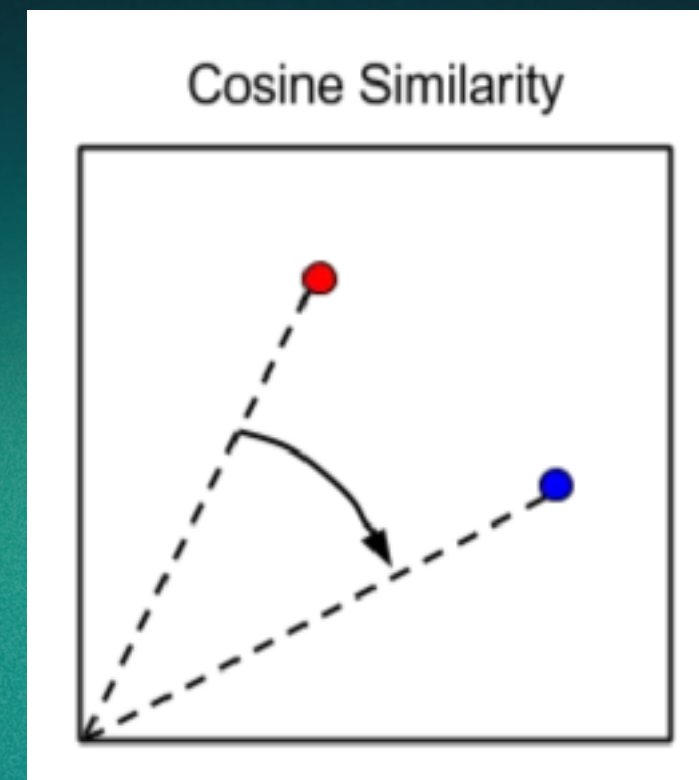
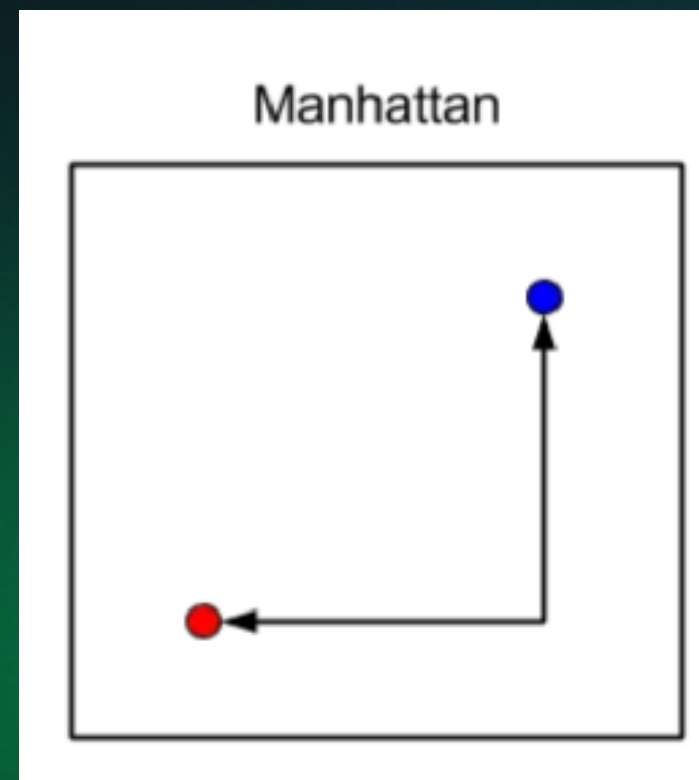
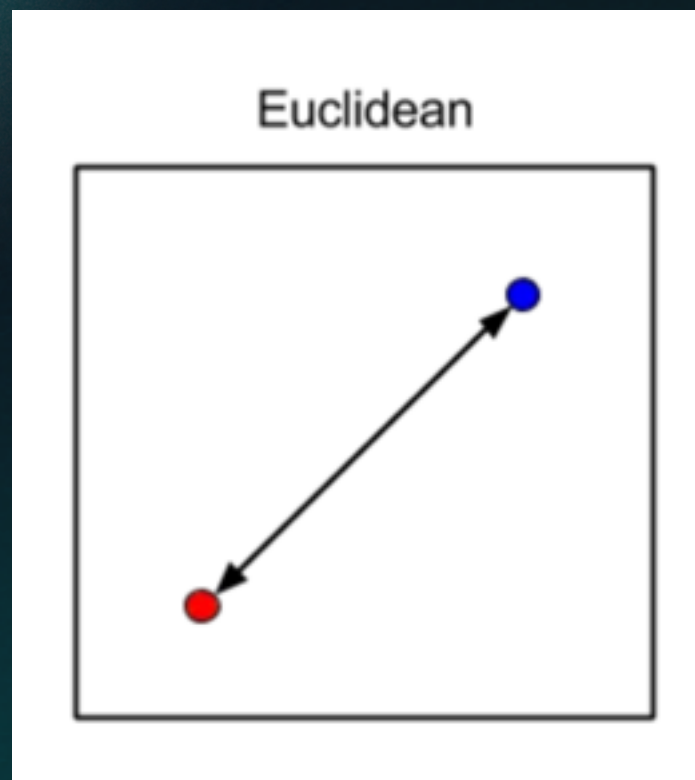
- Red Green Blue
- [255,101,206]
- #FF65CE
- [FF,65,CE]
- Hue Saturation Brightness
- [100,11,35]
- Combined Vector
- [[255,101,206],
[100,11,35]]





Color Demo

Measuring Distances



- Euclidean Distance
- Manhattan Distance
- Cosine Similarity
- Dot Product



Categorization Demo

Why use a vector?

AI Uses

- Natural Language Processing (NLP)
- Long Term memory for LLM
- Image/Unstructured Data Search

Other uses

- Similarity Search
- Recommendation Engines
- Semantic Searches
- Anomaly Detection



Search/Cache Demo

What are some Vector Databases?

Vector First

- Pinecone
- Chroma
- Weaviate

Vector search options

- Couchbase
- Postgres
- Redis

What to look for in a Vector Database?

Features to compare

- Performance
- Scalability
- Cost
- Maturity
- Reliability
- Indexing Techniques (HNSW, LSH, ENN, ANN)
- Vector Distance Calculation
- Vendor/Community
- AI Tool Integration

Credits and further learning

- John Strabla (Principle Product Manager @ Couchbase) - <https://github.com/jon-strabala>
- Matthew Groves (DevRel @ Couchbase) - <https://github.com/mgroves/>
- Couchbase GitHub - <https://github.com/couchbase-examples/hybrid-search-demo>
- Couchbase YouTube - <https://www.youtube.com/@CouchbaseInc>
- AssemblyAI - <https://www.youtube.com/watch?v=dN0lsF2cvm4>
- Fireship - <https://www.youtube.com/watch?v=kITvEwg3oJ4>
- PineCone - <https://www.youtube.com/@pinecone-io>
- Chroma - <https://www.trychroma.com/>
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Moonbeam

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9	3	8	5	8	6	6	2	6	8	9	9	3	0	5	4	3	7	8	3	3	5	1	4
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9	0	2	4	6	5	2	0	0	2	2	2	3	5	5	1	7	0	5	1	7	8	1	8

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Thank You!

The Future of **Modern Development** Starts Here, with you! 🚀